

Reproducing and Extending TISER: Context–Memory Conflict and Tennis-Domain Adaptation for Temporal Reasoning

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Abstract—Temporal reasoning requires a model to organise events before answering questions about order, duration, and overlap. We reproduce TISER, a framework that elicits a structured reasoning trace with an explicit timeline and reflection step, and evaluate two extensions with different goals. First, an inference-time Context–Memory Conflict probe tests whether a model follows supplied evidence when it contradicts a fact the model recalls without context. Second, a supervised tennis-domain study tests whether TISER transfers to a new type of temporal narrative and whether further adaptation helps. The reproduction reaches macro exact match of 0.878 and macro F1 of 0.949 on five in-domain benchmarks. On 1,056 genuine conflict cases, combining the TISER-trained model and prompt raises context-faithful exact match from 0.348 to 0.774, although reflections rarely acknowledge the context–memory conflict explicitly, and answer faithfulness remains low for order reversals. On the 113-record tennis holdout, continued adaptation raises exact match from 0.575 to 0.735. Original-domain retention is inconclusive under the predefined margin, so the conditional replay branch is not triggered. These results distinguish grounded use of context from explicit self-auditing and show that temporal-reasoning behaviour can transfer across domains.

I. Introduction

Temporal reasoning—interpreting event order, overlap, and duration—is central to question answering and narrative understanding. Large language models often identify the relevant events yet misalign them in time, confuse start and end points, or return a semantically correct answer in a form that exact-match metrics reject.

TISER [1] addresses this problem by training a model to produce four tagged stages before its final response: <reasoning>, <timeline>, <reflection>, and <answer>. We first reproduce its in-domain evaluation and then study two distinct extensions:

- 1) Context–Memory Conflict is an inference-time robustness probe. It leaves model weights fixed, changes the evidence supplied at evaluation time, and asks whether reflection exposes a disagreement between that evidence and recalled knowledge.
- 2) Tennis-domain adaptation is a supervised transfer experiment. It introduces a new domain and updates LoRA adapters using tennis temporal-QA traces.

The distinction is therefore both methodological and interpretive: the first extension measures robustness without

learning, while the second measures what transfers and what improves through additional learning.

Our contributions are a full TISER reproduction, a controlled conflict benchmark with answer-preserving controls, and a cross-domain adaptation and retention study at two model scales. Source code and run records are available in the project repository.

II. Background and Problem Definition

A. TISER Formulation

Each temporal-QA example consists of a question q , temporal context c , and gold answer a . TISER trains the model to generate a structured sequence $y = (r, t, s, a)$, where r is initial reasoning, t is an explicit timeline, s is a self-reflection, and a is the final answer. During evaluation, only the extracted <answer> span is scored.

The reproduction covers TGQA [2], TempReason L2 and L3 [3], and TimeQA easy and hard [4]. Out-of-distribution ToT-semantic examples are reported separately and excluded from the five-split in-domain macro average.

B. Research Questions

The reproduction asks whether our training pipeline recovers the original paper’s in-domain performance regime. Extension 1 asks three questions: RQ1, do the TISER prompt and fine-tuning improve adherence to context when memory disagrees; RQ2, does the reflection explicitly identify that disagreement; and RQ3, does the answer depend on conflict type? Extension 2 asks whether TISER transfers to tennis narratives, whether continued tennis adaptation improves over direct transfer, and whether the adapted model retains its original-domain performance sufficiently to avoid replay training.

III. Methodology

A. Baseline Reproduction

We use the released TISER supervision format and train a Qwen2.5 instruction model [6] to produce reasoning, timeline, reflection, and answer tags autoregressively in one pass. Prompt tokens are excluded from the loss so supervision targets only the structured completion. At

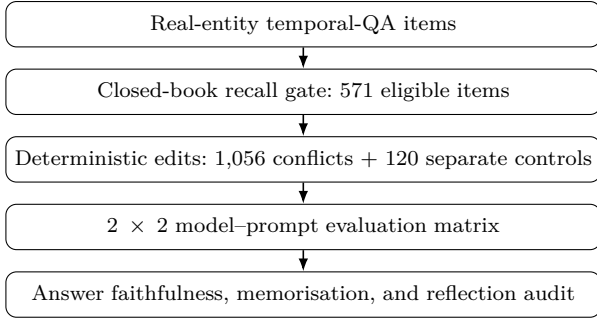


Fig. 1. Context-Memory Conflict is an inference-only evaluation pipeline. Controls preserve the answer and are kept separate from genuine conflicts.

evaluation time, the answer tag is extracted, normalized, and scored using Exact Match (EM) and token-level F1. Section IV contains implementation and hyperparameter details.

B. Extension 1: Context-Memory Conflict

a) Memory elicitation and eligibility.: We begin with real-entity TempReason L2/L3 and TimeQA easy/hard items; TGQA is excluded because its fictional entities do not provide the parametric beliefs needed for this probe. For each item, the context is removed and five closed-book answers are sampled. The majority answer m is retained when it occurs at least three times, and an item is eligible only if m equals the original gold answer. This yields 571 items for which the model demonstrates a stable, correct recalled belief.

b) Conflict construction.: For each eligible item, deterministic edits produce one or more of three genuine conflict classes: C1 shifts a date range so another entity covers the queried time; C2 replaces the answer entity with a same-type distractor; and C3 reverses intervals so a before/after relation changes. Each conflict must satisfy $\text{answer}(c') \neq m$. The procedure creates 1,056 genuine conflicts: 203 C1, 520 C2, and 333 C3. Separately, 120 controls reorder context rows while preserving the answer, so $\text{answer}(c') = m$. Controls are analysed as a sanity check and are not counted as conflicts.

c) Run matrix and measures.: We cross model weights $\in \{\text{vanilla}, \text{TISER-trained}\}$ with prompt $\in \{\text{standard}, \text{TISER}\}$, producing a 2×2 matrix without further training. Context-faithful EM/F1 measures agreement with $\text{answer}(c')$; memorised EM measures agreement with m ; malformed rate records unusable output. For TISER-prompt runs, the reflection audit has two stages: Judge A and Judge B independently label each <reflection>, after which an adjudicator resolves disagreements. A faithful answer without an explicit conflict label is a silent override.

C. Extension 2: Tennis-Domain Adaptation

a) Dataset creation and processing.: The tennis collection is synthetic and was created with ChatGPT-

5.5 in two stages using the prompts in Appendix A. First, ChatGPT-5.5 generated raw context-question-answer triples from scratch using the raw-example prompt. Second, ChatGPT-5.5 received training examples together with their gold answers and generated supervised TISER reasoning, timeline, reflection, and answer traces using the trace-generation prompt. The examples were generated synthetically rather than drawn from an external tennis corpus. The dataset is released under CC BY 4.0; repository code is released under MIT.

The recorded processing pipeline audits the schema, assigns identifiers and rule-based categories, converts records to standard and TISER-compatible prompts, checks duplicates, and performs a category- and exact-duplicate-group-aware 70/10/20 split with seed 42. The three disjoint data files contain 785 training examples, 113 examples in `tennis_dev.json`, and 224 examples in `tennis_test.json`, jointly covering all 1,122 records. In the reported experiments, the 224-record test file serves as the model-selection split, while the 113-record development file is reserved as the final holdout. Trace generation was restricted to the training split. Of the 600 traces used for training, 200 lacked the requested tags and were wrapped around their generated reasoning and supplied gold answer; 26 records were also recovered from 19 malformed JSONL lines. These repairs restored file structure rather than validating temporal correctness, which was assessed separately by the trace audit. The conversion CLI’s reported mode reproduces the processed files; strict-XML mode is reserved for new experiments.

b) Adaptation strategies.: At 0.5B scale, we compare the vanilla model under standard and TISER prompts with a tennis-only adapter. At 7B scale, we first test direct transfer of the original TISER adapter and then continue LoRA training from that adapter on tennis traces. The evaluator retains TISER’s output parser while adding tennis-specific normalization for rankings, minute durations, punctuation, and articles.

IV. Experimental Setup

A. Data and Evaluation

After a 2,048-token filter, TISER training uses 54,488 examples. The test set has 19,214 in-domain examples and 2,800 ToT-semantic OOD examples. The conflict subset is constructed from 15,898 real-entity in-domain items and is evaluated on the 1,056 conflicts and 120 controls described above.

B. Tennis Split and Trace Accounting

The 785-record training split is the candidate pool for supervised trace creation; it is not itself the file used for optimization. It partitions exactly as $50 + 600 + 135 = 785$. Train positions 0–49 have valid traces from an initial pilot and support only pilot/smoke runs. Train positions 50–649 are the disjoint 600-record trace collection used by both reported tennis adaptations. For positions 650–784,

requests were prepared in batches 13–15 but no generated outputs are present, so those 135 examples could not be used as supervised trace targets. The repository therefore contains traces for 650 unique training records overall, but the reported models use only the 600-record full-run file.

The 113 examples in `tennis_dev.json` have no training traces, were never used for gradient updates, and were excluded from model selection for use as the final holdout. The 224 examples in `tennis_test.json` likewise have no training traces and never affect gradients, but they were used to evaluate every completed condition and rank the 7B candidates; they therefore serve as the selection split. All selection evaluations use the same 224 IDs in the same order. The complete 785-record split and unused generation requests document the deterministic 1,122-record partition, pilot provenance, and trace coverage.

C. Tennis Semantic and Trace Audits

Structural validity does not establish that a gold answer is uniquely entailed by its context or that a generated trace supports its answer. We therefore completed two UI-mediated, blinded file-batch audits. The semantic audit covers 1,121 unique context–question–gold payloads, mapping to all 1,122 source rows because two rows are identical. The trace audit covers all 650 available training traces: the 50-record pilot and the separate 600-record collection used for the reported adaptations. Semantic labels are supported, wrong gold, underdetermined, inconsistent, and unscorable; trace labels are supported, unsupported reasoning, contradictory, and unscorable.

For each audit, Judge A and Judge B ran in separate Codex tasks [8], configured in the UI as `gpt-5.6-sol` with high reasoning effort. Each task received only its assigned bundle, pass-specific rubric, and the common task prompt reproduced in Appendix B. Every scorable decision required item-specific reasoning and an exact quotation from the relevant source field. Hashes associate each response with its batch, item, source, and prompt. The two passes disagreed on 9 semantic items and 12 traces. A third task adjudicated every disagreement; the semantic workflow also adjudicated every proposed gold-answer change, producing 12 semantic and 12 trace adjudications in total. All expected decisions passed schema, identity, and exact-evidence validation.

These audits assess the data used to train the reported adapters; they do not alter the training inputs. Semantic adjudication produces alternative scoring views while preserving the original source files. The Codex UI did not expose a verifiable backend snapshot or decoding settings, so these details are unavailable. All primary and adjudication decisions used the same displayed model label, and no human calibration was performed.

D. Models, Training, and Decoding

Table I consolidates the training hyperparameters. LoRA [5] targets the q , k , v , and o attention projections.

TABLE I
Hyperparameters for the reported trained adapters. Micro. $\times$ GA gives the per-device microbatch and gradient-accumulation factor.

Run	n	Ep.	Updates	Micro. $\times$ GA	Quant.
TISER 7B	54,488	3	10,215	8 $\times$ 2	none
Tennis 0.5B	600	1	600	1 $\times$ 1	none
Continued 7B	600	2	74	4 $\times$ 4	4-bit NF4

Optimisation Paged 8-bit AdamW; learning rate 2×10^{-4} ; cosine schedule; warmup ratio 0.03; weight decay 0; sequence limit 2,048; seed 42.

LoRA Rank 16; $\alpha = 32$; dropout 0.05; targets q , k , v , and o projections; no bias.

Checkpointing Enabled for TISER 7B and continued 7B; disabled for tennis 0.5B. The 4-bit run uses double quantisation.

The reproduction uses completion-only loss and no base-model quantization; the 7B continued-adaptation run loads the base in 4-bit precision. All reported answer generations are greedy. Reproduction and conflict runs allow up to 768 new tokens; tennis evaluation allows up to 768 tokens at 0.5B and 256 at 7B.

The continued-adaptation condition was selected from a grid over one to three epochs, learning rates $\{5\times 10^{-5}, 1\times 10^{-4}, 2\times 10^{-4}\}$, and gradient accumulation $\{4, 8\}$ with per-device batch size 4. All 22 candidates were evaluated on the same 224-record selection split, enabling direct comparison. Continued adaptation achieved the highest selection-set score and led two runners-up by one correct answer. Its fixed-protocol rerun reaches 0.728 EM and 0.852 F1 on this split. The selected adapter was then kept unchanged and evaluated once on the separate 113-record final holdout.

The retention experiment defines up to five conditions. C0 is the original TISER adapter, and C1 is the selected tennis-adapted model. If the retention gate detects forgetting, C1R is a fresh current-environment reproduction trained on the same 600 tennis traces; R25 is the replay condition trained on those traces plus 200 original-TISER examples; and R25-T is a replay sensitivity condition matched as closely as possible to C1R’s supervised-token exposure.

The protocol first compares C0 and C1 on a predefined seed-42 sample of up to 100 items per original-TISER split. A 10,000-resample paired interval for the five-split macro-EM difference triggers C1R and R25 training only when its upper bound is below -0.02 ; a lower bound above -0.02 stops as non-inferior, and other intervals stop as inconclusive. If triggered, C1R and R25 both start from C0 and run for 74 updates under the same current environment. R25-T is run only when R25 supervised-token exposure differs from C1R by more than 10%. The completed semantic audit retains the original 113-record final-holdout view from `tennis_dev.json` and the 224-record selection view from `tennis_test.json`, without proposed changes or exclusions. The final campaign eval-

uates the applicable fixed conditions on the 113 tennis inputs; original-domain retention is reported from the predefined sample. The reflection and trace audits are independent of this evaluation. Its results are not used for further model selection or training.

E. Conflict Sampling and Reflection Audit

Closed-book recall uses five samples with temperature 0.7 and top- p 0.9; the subsequent run matrix uses one greedy decode per item. We use item-level bootstrap confidence intervals with 2,000 resamples and seed 42, plus paired McNemar tests for the two headline axes.

The earlier Claude-based aggregate is excluded from the present results because its exact model version, complete prompt, and row-level decisions are unavailable. We instead conducted a UI-mediated audit of the saved outputs. The audit deduplicates the two 1,176-row TISER-prompt cells into 2,295 non-empty reflections, which map back to 2,348 source rows; four malformed source rows are retained as unscorable. Judge A and Judge B ran in separate Codex tasks configured in the UI as gpt-5.6-sol with high reasoning effort. Each task received only its blinded reflection bundle and a conservative pass-specific rubric, without model condition, conflict class, answer correctness, performance, or the other pass’s decisions. Responses required an item-specific rationale and an exact source substring for every positive label. Hashes associate the evidence with its batch, item, source, response, and prompt; the shared judge prompt is reproduced once in Appendix B.

The primary decision is one of three labels: explicit conflict, no explicit conflict, or unscorable. A positive label requires the reflection itself to acknowledge tension between supplied context and prior knowledge, memory, expectation, or an alternative factual answer; routine timeline checking is negative. The two passes disagreed on 66 of 2,295 reflections. A third Codex task, also configured as GPT-5.6 Sol with high reasoning, adjudicated those items from the source reflection and both primary decisions under the same conservative rule. All 66 adjudications passed the predefined schema and exact-evidence checks.

F. Compute and Software

The reproduction was trained on one H200 SXM GPU in bf16 for approximately 6.5 hours. Test and conflict generation used vLLM [7] on an H100 SXM GPU; the full 22,014-example evaluation took approximately 17 minutes. Reported versions were PyTorch 2.4.1/CUDA 12.4, Transformers 4.46.3, PEFT 0.13.2, and TRL 0.12.2 for reproduction, and vLLM 0.22.0 with PyTorch 2.11/CUDA 13.0 for accelerated inference.

V. Results

A. Baseline Reproduction

Table II reports the full in-domain evaluation. Our macro-EM is 0.033 below the paper, while macro-F1 is

TABLE II
TISER reproduction on five in-domain splits. Paper is the comparable Qwen2.5-7B+TISER result; ToT-semantic (OOD EM 0.381) is excluded from the macro.

Split	n	Ours		Paper	
		EM	F1	EM	F1
TGQA	3,316	0.571	0.870	0.845	0.942
TempReason L2	5,397	0.907	0.929	0.855	0.875
TempReason L3	4,426	0.961	0.974	0.915	0.949
TimeQA easy	2,997	0.980	0.989	0.979	0.983
TimeQA hard	3,078	0.970	0.981	0.961	0.972
Macro (5)	19,214	0.878	0.949	0.911	0.944

TABLE III
Run matrix. Conflict metrics use only genuine conflicts ($n = 1,056$); control EM is separate ($n = 120$). Δ EM is relative to base+standard on conflicts.

Condition	Faith-EM	Faith-F1	Mem-EM	Δ EM	Ctrl. EM
Base + standard	0.348	0.579	0.295	—	0.658
Base + TISER	0.525	0.718	0.228	+0.176	0.883
TISER + standard	0.560	0.759	0.251	+0.211	0.700
TISER + TISER	0.774	0.914	0.153	+0.425	0.900

0.005 higher; the reproduction is therefore in a similar overall performance regime, but its error profile differs substantially on TGQA.

TGQA combines low EM (0.571) with much higher F1 (0.870), a pattern consistent with partially correct or differently formatted answers. The repository does not preserve the row-level annotations needed to substantiate a previously reported adjusted-accuracy estimate, so we omit that estimate rather than treating it as a reproducible result.

B. Context–Memory Conflict

Table III makes the 1,056 genuine conflicts the primary population and reports the 120 controls separately. The TISER-trained model with the TISER prompt reaches faithful-EM 0.774, compared with 0.348 for the vanilla model under a standard prompt. Both interventions help: relative to the same-model standard prompt, TISER prompting adds 0.214; relative to vanilla weights under the same TISER prompt, TISER training adds 0.249.

For continuity with the original aggregation, faithful-EM over conflicts plus controls is 0.380, 0.561, 0.574, and 0.787 in table order. The previously reported rise from 0.380 to 0.787 therefore includes controls; the conflict-only comparison is 0.348 to 0.774. On all 1,176 constructed rows, the prompt-axis difference is +0.213 and the model-axis difference is +0.225; both paired McNemar tests give $p < 10^{-40}$.

Figures 2 and 3 separate the model–prompt comparison from the conflict-type analysis.

Context-faithful exact match		
	Prompt format	
	Standard	TISER
Model	Base model	0.380 [0.352, 0.407]
	TISER SFT	0.574 [0.546, 0.603]
		0.561 [0.531, 0.589]
		0.787 [0.761, 0.810]

Value [95% bootstrap CI]; $n = 1,176$ per cell

Fig. 2. Context-faithful exact match across the 2×2 model-prompt matrix. Each cell reports EM and its item-level bootstrap 95% confidence interval over all 1,176 evaluation rows, including 120 answer-preserving controls. Both the TISER prompt and TISER fine-tuning improve faithfulness to the edited context.

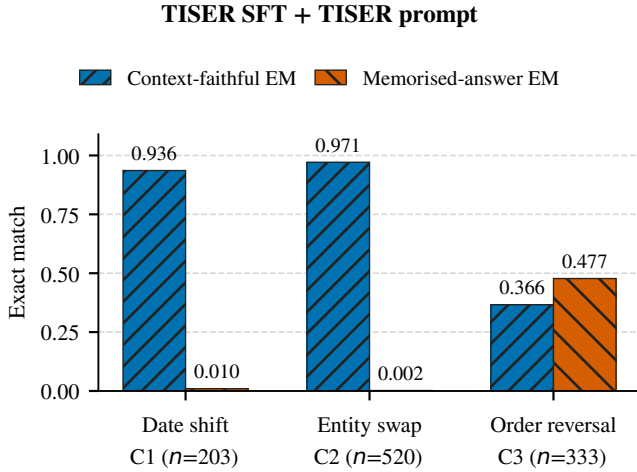


Fig. 3. Faithful and memorised-answer exact match for the TISER-fine-tuned model with the TISER prompt, shown only for the three genuine conflict classes (controls excluded). Date shifts and entity swaps are overwhelmingly resolved using the edited context; order reversals are the only class in which memorised-answer EM exceeds context-faithful EM.

C. Conflict Type and Reflection

Table IV isolates the three genuine conflict types for the strongest condition. Date shifts and entity swaps are largely solved, but order reversal is a clear blind spot: it is the only class for which memorised EM exceeds faithful EM.

The replacement audit finds explicit conflict acknowledgement in 87/1,053 scorable genuine conflicts for the vanilla model with the TISER prompt (8.26%, Wilson 95% CI 6.75–10.08%) and 36/1,056 for the TISER-trained model with the TISER prompt (3.41%, 2.47–4.68%). Control false-positive rates are 3/120 (2.50%, 0.85–7.09%) and

TABLE IV
Genuine-conflict results for TISER model+prompt. Refl. is the secondary LLM-judge conflict-naming rate.

Class	n	Faith-EM	Mem-EM	Refl.
C1 date shift	203	0.936	0.010	0.020
C2 entity swap	520	0.971	0.002	0.006
C3 order reversal	333	0.366	0.477	0.087
All conflicts	1,056	0.774	0.153	0.034

TABLE V
Completed GPT-5.6 Sol reflection audit. Conflict and control columns report explicit flags; disagreement compares the two primary passes; silent counts faithful genuine-conflict answers without a flag. Denominators exclude three unscorable genuine-conflict reflections for Base+TISER and one unscorable control for TISER+TISER.

Condition	Conflict	Control	Disagree	Silent
Base + TISER	87/1,053	3/120	41/1,173	543/554
TISER + TISER	36/1,056	3/119	25/1,175	810/817

3/119 (2.52%, 0.86–7.15%), respectively. In the strongest answer-faithfulness condition, conflict flags remain concentrated in C3: 4/203 C1 items (1.97%), 3/520 C2 items (0.58%), and 29/333 C3 items (8.71%). Among its 817 faithful genuine-conflict answers, 810 have no explicit flag, yielding a 99.14% silent-override rate (98.24–99.58%). The corresponding rate for the vanilla model with the TISER prompt is 543/554, or 98.01% (96.48–98.89%). The 3.41% genuine-conflict rate is descriptively close to the 2.52% control rate, so these results support rarity and C3 concentration but do not establish reliable discrimination between genuine conflicts and controls. The separate control arm reaches faithful-EM 0.900 in the strongest condition; because its answer equals memory by construction, it measures preservation of answerability rather than conflict resolution. A silent override remains a behavioural description, not proof of an internal conflict-resolution mechanism.

D. Tennis-Domain Adaptation

Table VI reports all completed 224-example selection-set runs. At 0.5B, prompting alone does not help, while tennis-only training gives a modest gain. At 7B, the original TISER adapter transfers more strongly, and the fixed-protocol rerun of continued adaptation adds 0.147 EM and 0.151 F1. All 22 candidates were evaluated on the same 224-record selection split, enabling direct comparison. Continued adaptation achieved the highest selection-set score.

E. Retention and Final Holdout

The retention evaluation sampled 100 records from each of six original-TISER splits. The primary comparison uses the five in-domain splits ($n = 500$); ToT-semantic remains separate OOD evidence. C0 reaches 0.880 macro-EM and C1 reaches 0.876, a difference of -0.004 with a paired 95%

TABLE VI

Tennis temporal QA on the 224-record selection split stored in tennis_test.json. Deltas are against the first row within each model-size block; Malformed is an output count.

Condition	EM	F1	Δ EM	Δ F1	Malf.
Qwen2.5-0.5B-Instruct					
Base, standard prompt	0.379	0.472	—	—	0
Base, TISER prompt	0.375	0.420	-0.004	-0.052	5
Tennis-only adapter	0.464	0.516	+0.085	+0.045	0
Qwen2.5-7B-Instruct					
Original TISER	0.580	0.701	—	—	0
Continued adaptation	0.728	0.852	+0.147	+0.151	0

TABLE VII

Original-domain retention and final tennis-holdout results. Differences are C1-C0; intervals use 10,000 paired bootstrap resamples with seed 42.

Evaluation	C0	C1	Δ	95% CI
Retention macro EM	0.880	0.876	-0.004	[-0.028, 0.020]
Retention macro F1	0.946	0.936	-0.010	[-0.028, 0.007]
Tennis holdout EM	0.575	0.735	+0.159	[0.080, 0.239]
Tennis holdout F1	0.677	0.834	+0.157	[0.082, 0.232]

interval of $[-0.028, 0.020]$. This interval satisfies neither the clear-forgetting nor the non-inferiority rule, so the gate is inconclusive and C1R, R25, and R25-T are not trained. On the separate 100-record ToT-semantic sample, EM changes from 0.410 to 0.320 and F1 from 0.410 to 0.347; this split does not enter the gate.

The final tennis evaluation uses all 113 audited holdout records. Continued adaptation raises EM from 0.575 to 0.735 and F1 from 0.677 to 0.834. The paired EM difference is 0.159 (95% CI 0.080–0.239), and the exact McNemar test gives $p = 0.000277$. Table VII reports the paired summaries.

F. Tennis Audit Results

Table VIII summarizes the completed two-pass audits. The semantic decisions map back to all 1,122 source rows: all 113 final-holdout records in tennis_dev.json and all 224 selection records in tennis_test.json are model-judge supported, so both evaluation views retain their original labels and denominators. All 18 semantic concerns occur in training: 13 records are underdetermined, four have a model-adjudicated proposed gold-answer change, and one is internally inconsistent. Fourteen of these records belong to the reported 600-record training set and four to the untraced tail; none belongs to the 50-record pilot. The four proposed changes are tennis_000704, tennis_000716, and tennis_000885 from “Yes” to “No,” and tennis_000948 from “Djokovic hit two aces” to “Sinner elected to receive.” These proposed changes were not substituted into the training files used for the reported runs.

The trace audit supports all 50 pilot traces. Within the 600 traces used for the reported adaptations, 582 are supported, 14 contain unsupported reasoning, and

TABLE VIII

Completed tennis model-judge audits. Label abbreviations are U: underdetermined; WG: wrong gold; I: inconsistent; UR: unsupported reasoning; C: contradictory.

Audit	n	Supported	Other final labels	A/B dis.
Semantic (unique)	1,121	1,103	13 U; 4 WG; 1 I	9
Trace	650	632	14 UR; 4 C	12

four are contradictory. Combining the semantic and trace decisions, 21 distinct records in the 600-record training set have at least one concern; 579 are supported by both model audits. A cross-audit comparison also finds six records for which the semantic and trace decisions imply different readings: three have a supported semantic record but an unsupported trace, while three have a flagged semantic record but a supported trace. This audit examines the training data used by the reported models rather than measuring the causal effect of individual records: that effect cannot be inferred without retraining or ablation, and the model-judge decisions are not human certification.

VI. Discussion and Limitations

The reproduction is strong overall but exposes the sensitivity of EM to answer formatting. More importantly, the two extensions answer different questions. The conflict probe shows that training and prompting can improve use of supplied evidence without making reflection an explicit contradiction detector. The tennis study uses different starting conditions at 0.5B and 7B, so the two comparisons are interpreted separately rather than as a controlled test of model scale.

Several limitations constrain these conclusions. The conflict set is restricted to facts the model already recalls confidently, so it is not a random sample of temporal questions. The completed reflection, semantic, and trace audits use two primary passes and adjudication, but all judgments use the same UI-selected model family. The exact backend snapshot and decoding settings are unavailable, and no human calibration establishes judge accuracy or shared-bias rates. Greedy decoding provides one subject-model output per condition. For tennis, the audits identify concerns in 21 distinct records inside the 600-record training set, whose size reflects an operational cutoff rather than a principled sample size; their causal effect remains unmeasured. The best 7B setting was selected on the 224-record selection split and evaluated once on the separate 113-record holdout. The retention interval establishes neither clear forgetting nor non-inferiority under the predefined margin. Consequently, the conditional replay branch was not executed, and this study makes no claim about replay effectiveness. Training variability remains unmeasured because each adapter represents one training run.

VII. Conclusion

We reproduced TISER’s strong in-domain temporal-QA performance and used two extensions to test complementary aspects of generalisation. Structured prompting and training made the model more willing to follow conflicting context. Its reflections rarely acknowledged the conflict explicitly, and its answers remained weak on reversed event order. On the 113-record final holdout, continued tennis adaptation improved EM from 0.575 to 0.735 over direct transfer from the original TISER adapter. Original-domain retention was inconclusive under the predefined margin, so the conditional replay branch was not executed. Overall, following evidence, explaining a conflict, and adapting to a new domain are distinct capabilities and should be evaluated separately.

VIII. Reproducibility

The repository provides command-line workflows for data preparation, training, evaluation, scoring, auditing, and statistical analysis. It records canonical configurations, seed 42, resolved settings, predictions, metrics, and available environment metadata for the reported models and results. Dataset provenance, licences, generation prompts, and trace coverage are documented separately.

The audit records retain the shared task prompt, pass-specific rubrics, original responses, hashes, decisions, adjudications, row-level outputs, and summaries. The reflection, semantic, and trace audits contain 4,590/66, 2,242/12, and 1,300/12 primary judgments/adjudications, respectively; the semantic audit also provides the 113- and 224-record evaluation views.

The Colab notebook implements the conditional retention and final-evaluation workflow with resumable stages. The completed branch evaluates C0 and C1 on the predefined retention sample, records the inconclusive gate without starting replay training, and evaluates both fixed adapters on the 113-record tennis holdout. File hashes associate each run with its data, configuration, adapter, and scoring view.

Appendix A

Tennis Data-Generation Prompts

The tennis data were created with ChatGPT-5.5 in two stages using the prompts below.

A. Stage 1: Raw Example Creation

You are creating a synthetic domain-adaptation dataset for temporal reasoning in the professional tennis domain.

The dataset will be used to adapt a TISER-style temporal reasoning model. Your task is to generate RAW examples only.

For each example, output exactly one JSON object with these fields:

```
{
  "context": "...",
  "question": "...",
  "answer": "..."
}
```

Do NOT generate reasoning.
Do NOT generate timeline.
Do NOT generate reflection.
Do NOT add explanations.
Do NOT wrap the output in markdown.
Do NOT use code fences.

Return a JSON array containing exactly 50 examples.

GENERAL GOAL

Each example must test temporal reasoning in tennis match or tournament narratives.

The model should need to reason about the order, overlap, duration, or sequence of events. The answer must be inferable only from the context.

DOMAIN

Use professional tennis scenarios such as:

- match progression
- first set, second set, third set
- breaks of serve
- service holds
- tie-breaks
- set points
- match points
- rain delays
- roof closure
- medical timeouts
- injury treatment
- warm-up interruptions
- tournament rounds
- quarterfinals, semifinals, finals
- player withdrawals
- comebacks
- coaching violations
- crowd interruptions
- changeovers

You may use realistic player names such as Sinner, Alcaraz, Djokovic, Medvedev, Zverev, Rune, Rublev, Fritz, Nadal, Federer, Tsitsipas, Ruud, Shelton, De Minaur.

The events do NOT need to be historically true. Treat them as fictional but realistic tennis narratives.

CONTEXT REQUIREMENTS

Each context must:

- contain 3 to 6 sentences
- contain at least 3 temporal events
- be self-contained
- be unambiguous
- avoid external knowledge
- use natural sports-report language

- include clear temporal signals such as:

- before
- after
- during
- while
- later
- earlier
- immediately after
- at the same time
- minutes later
- in the first set
- in the second set
- before the tie-break
- after the medical timeout

QUESTION REQUIREMENTS

Generate a balanced mix of question types:

1. Yes/No temporal comparison

Example:

"Did Sinner call the medical timeout before breaking serve?"

2. Which happened first?

Example:

"Which happened first: Alcaraz saved a break point or Medvedev won the tie-break?"

3. Which happened last?

Example:

"Which event happened last?"

4. What happened before X?

Example:

"What happened before the rain delay?"

5. What happened after X?

Example:

"What happened after Djokovic lost the second set?"

6. Duration / time gap

Example:

"How many minutes after the match started did the rain delay begin?"

7. During / overlap

Example:

"Was the roof being closed while the players were off court?"

8. Tournament progression

Example:

"Did Sinner reach the semifinal before playing the final?"

ANSWER REQUIREMENTS

Each answer must:

- be short
- be exact
- be unambiguous
- directly answer the question
- not include explanations

Use answer formats like:

- "Yes"
- "No"
- "Sinner broke serve"
- "The rain delay"
- "Alcaraz won the second set"
- "45 minutes"
- "The semifinal"

IMPORTANT BALANCE CONSTRAINTS

Among the 50 examples, approximately include:

- 15 Yes/No before-after questions
- 10 first/last ordering questions
- 8 what happened before/after questions
- 7 duration or time-gap questions
- 5 during/overlap questions
- 5 tournament progression questions

Difficulty distribution:

- 15 easy examples with direct before/after reasoning
- 25 medium examples with 3-4 ordered events
- 10 hard examples with 5-6 events or distractor events

QUALITY RULES

Avoid ambiguous cases such as:

- "in 2014" without month/day when asking about a specific month
- unclear transitions between teams, rounds, or sets
- answers that require real tennis knowledge
- questions with multiple valid answers
- contexts where two events could reasonably be interpreted in different orders

Do not repeat the same structure too many times.

Do not generate near-duplicate examples.

Do not always use the same players.

Do not always make the answer "Yes".

Use both positive and negative examples.

Before producing the final output, internally check each example:

- Is the answer derivable from the context?
- Is the temporal order clear?
- Is there only one correct answer?
- Is the answer short?
- Is the example about tennis?
- Did you avoid reasoning/timeline/reflection?

OUTPUT FORMAT

Return only a valid JSON array.

Example format:

```
[
  {
    "context": "Sinner held serve in the opening game. Djokovic called a medical timeout before Sinner broke serve in the fourth game. After the break of serve, rain delayed the match for twenty minutes.",
    "question": "Did Djokovic call the medical timeout before Sinner broke serve?",
    "answer": "Yes"
  },
  {
    "context": "Alcaraz saved two break points at 2-3 in the first set. Medvedev broke serve three games later. The set ended with Medvedev winning a tie-break.",
    "question": "Which happened first: Medvedev broke serve or Alcaraz saved two break points?",
    "answer": "Alcaraz saved two break points"
  }
]
```

Now generate exactly 50 examples.

This prompt was used to generate the 1,122 raw context-question-answer records before schema auditing, categorisation, and splitting.

B. Stage 2: TISER Trace Generation

You will receive tennis temporal reasoning training examples.

For each example, generate only one JSONL record with the following fields:

```
{
  "question_id": "...",
  "output": "..."
}
```

The output must follow the TISER format:

```
<reasoning>
...
<timeline>
...
</timeline>
<reflection>
...
</reflection>
...
</reasoning>
<answer>GOLD_ANSWER</answer>
```

Rules:

- Use only information contained in the provided context.
- Reason explicitly about the temporal relations required to answer the question.
- Construct a chronological timeline containing the relevant events.
- In the reflection, verify that the reasoning and timeline support the gold answer.
- Do not invent events, dates, entities, or temporal relations.
- The final answer must match the provided gold_answer exactly.
- Preserve the question_id exactly.
- Output one JSON object per line.
- Output JSONL only. Do not include markdown, explanations, headings, or any additional text.

The requests also supplied question_id, category, context, question, and gold answer. Generated rows contain question_id and output. The final 600 records exactly match batches 1–12 in order. Nineteen malformed physical JSONL lines covering 26 records were repaired, and 200 outputs that lacked the requested XML-like tags were mechanically wrapped using their existing text and supplied gold answer. All 600 final rows pass structural and gold-answer-equality checks; those checks do not establish semantic entailment.

Appendix B

Model-Judge Prompt and Adjudication

The reflection, semantic, and trace audits used the same prompt below, byte for byte. It was pasted unchanged into the separate Judge A, Judge B, and adjudicator Codex tasks for each audit. Each task also received only its assigned JSON bundle, which carried the audit-specific, pass-specific rubric and blinded items.

Audit the attached JSON bundle as one independent model-judge pass.

Use only the attached bundle as evidence. Do not inspect any repository, mapping, other judge pass, previous decision, historical result, or experiment output. Read the rubric in each batch and judge every item individually. Do not use scripts, regular expressions, keyword rules, lexical heuristics, or a default label to make judgments. Code may only assemble and validate JSON after the item-specific judgments have been made.

Process every batch in the bundle. For each batch, create one response file named BATCH_ID.json in a single new response directory, replacing BATCH_ID with the batch_id value. Each response file must contain exactly:

```
{
  "batch_id": "...",
  "batch_sha256": "...",
  "pass": "judge_a or judge_b or adjudicator",
  "judged_at": "timezone-bearing ISO 8601 timestamp",
  "observed_model_label": "gpt-5.6-sol",
  "reasoning_effort": "high",
  "judgments": [
    {
      "audit_id": "...",
      "source_sha256": "...",
      "label": "...",
      "corrected_answer": null,
      "mention_kind": "none",
      "evidence": [
        {
          "field": "context or trace or reflection",
          "quote": "exact substring"
        }
      ],
      "rationale": "brief item-specific explanation"
    }
  ]
}
```

Copy batch_id, batch_sha256, pass, audit_id, and source_sha256 without alteration. Include every item exactly once and use only labels allowed by that batch's rubric. Every rationale must explain its own item. Evidence quotes must be exact substrings of the named source field.

For a reflection labeled explicit_conflict, provide nonempty reflection evidence and set mention_kind to memory_context_mismatch, contradiction_or_inconsistency, uncertainty_or_doubt, or other. For no_explicit_conflict or unscorable, use an empty evidence list and mention_kind none. For semantic and trace judgments, mention_kind is always none and every label except unscorable requires evidence. Set corrected_answer to a nonempty replacement only for semantic wrong_gold; otherwise set it to null. Do not invent unavailable provenance.

Validate each response file before continuing. Preserve completed valid files if execution is interrupted and continue until the response directory contains exactly bundle.batch_count files covering every batch in bundle.batch_ids. Then report the response directory's absolute filesystem path.

A. Adjudication

Judge A and Judge B were compared by item identity. Follow-up bundles contained the original source and both decisions for every disagreement, plus every proposed wrong gold decision in the semantic audit. A third independent task, configured in the UI as gpt-5.6-sol with high reasoning effort, reapplied the relevant rubric without majority voting. The workflow adjudicated 66 reflection, 12 semantic, and 12 trace decisions. All 90 passed the label, identity, hash, and exact-quotation checks.

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